

1, Explain behaviour of hub device

Hub is a layer 1 device, It Lacks Intelligence like Switch, also known as Broadcast device. It is Half-Duplex meaning simultaneous transmission causes data Collision. When a data travels $A \rightarrow B$ and $C \rightarrow A$ at the same time Data collision happens.

2, Explain behaviour of switch device

Switch is an Intelligent Layer 2 device. It doesn't need to be configured. It does address learning and creates a MAC Table of the known port/devices.

When it knows the SM & DM it does Unicast, when it doesn't know the DM it does Broadcast.

The MAC Table contains Port no and MAC address.

3, Explain behaviour of router device

It is a layer 3 device. Used to connect different network devices. Router is used to break broadcast domain.

It contains Routing table consisting of Network ID and prefix to send the packet to different Network. ~~and~~ It needs to be configured.